PROJECT BRIEF

Learner-as-Designer: Teaching ISO/IEC 29119 (1-4) Through Game Design

ISO/IEC 29119, the international software testing standard, is often perceived as abstract and challenging to grasp. In this innovative project, you won't simply study the standard—you'll design an educational game to teach it to others. This approach follows the Learner-as-Designer pedagogical model: you learn most deeply when you create learning experiences for others. If you can teach a concept clearly and effectively, you truly understand it.

Group Structure

3 students per group working collaboratively across all project areas

Timeline

5-week structured progression from analysis to final presentation

Core Focus

Transform abstract ISO concepts into interactive, playable learning experiences

Why This Matters?

You're not just studying ISO/IEC 29119—you're reconstructing it and teaching it.

Project Objectives

By completing this project, you will achieve several interconnected learning outcomes. You'll develop a thorough understanding of ISO/IEC 29119's structure across Parts 1–4, identify genuinely difficult concepts within the standard, and analyse common misunderstandings that learners encounter. Beyond comprehension, you'll design an educational game aligned with ISO terminology, create structured visual explanations that clarify complex relationships, and experience the power of active, design-based learning firsthand.

WEEK-BY-WEEK GUIDE

Project Timeline & Deliverables

1 Week 1: Understand & Analyse

Deliverables: 2-page Concept Analysis + 1 Concept Map

Select a difficult ISO/IEC 29119 topic and analyse why it challenges learners. Identify where it appears in the standard, explore common misunderstandings, and create a structured diagram showing conceptual relationships. Goal: Achieve deep understanding before design begins.

2 Week 2: Game Design Proposal

Deliverables: 3-page Game Proposal + Rough Sketch

Propose your game design (card, board, digital, VR/AR, or LEGO-based). Define your target audience, establish 3–5 measurable learning objectives, explain how the game teaches the concept and reveals mistakes, and outline basic rules. Focus on learning first, game mechanics second.

3 Week 3: Prototype Version 1

Deliverable: Playable 10-minute prototype

Bring a testable version using authentic ISO terminology and clear rules. Other groups will test your game and provide structured feedback. Goal: Verify whether meaningful learning actually occurs during gameplay.

4 Week 4: Improvement & Visuals

Deliverables: Improved Prototype

Refine your game based on feedback and get new feedback from your teacher.

5 Week 5: Final Presentation

Deliverables: Final Game + Visuals + Documentation + Reflection

Present in 22 minutes: 5-minute explanation, 15-minute gameplay demonstration, 2-minute Q&A. Submit complete documentation and a 1-page reflection addressing what you understood better through design, the most difficult ISO concept encountered, and how design differs from memorisation.

Criterion	Points	Full Points
1. Concept Accuracy	20	Terminology correct. No conceptual errors. Clear explanation of relationships.
2. Learning Objectives	15	4–5 measurable objectives. Clear link between objectives and game mechanics.
3. Educational Effectiveness	20	Game requires reasoning. Misconceptions become visible.
4. Prototype Quality	35	Fully playable (15–20 min). Clear rules. Structured flow.
5. Reflection & Team Understanding	10	Clear insight on what was learned and why design helped.

Sample Learning Objectives for ISO/IEC 29119 (1–4)

After playing the game, participants will be able to:

1. **Define** key testing terms using correct ISO terminology.
2. **Differentiate** between product risks and project risks.
3. **Prioritize** test activities using risk-based reasoning.
4. **Distinguish** between test strategy and test plan.
5. **Design** basic test cases using appropriate test design techniques.
6. **Differentiate** between specification-based, structure-based, and experience-based techniques.
7. **Differentiate** between exploratory testing and scripted testing.
8. **Distinguish** between regression testing and retesting.
9. **Identify** required elements of a defect report.
10. **Determine** whether test completion criteria are satisfied.